**FYUGP B.Sc. PHYSICS HONOURS**

ELECTRODYNAMICS II

Semester V · Core in Major

**MODULE 3: ELECTROMAGNETIC WAVES**

Past questions

## Section A: 1 to 3 marks

***Waves in one dimension and the wave equation***

1 Write down the wave equation in one dimension. **(1 mark)** *April 2020*

***Polarisation and transverse waves***

2 State whether true or false: Electromagnetic wave is transverse and not longitudinal. **(1 mark)** *November 2019*

***Electromagnetic waves in vacuum***

3 State whether true or false: Electromagnetic waves travel with the speed of light in vacuum. **(1 mark)** *November 2021*

4 Give the wave equation for the magnetic field vector $\vec{B}$ in free space and explain the terms involved. Write down the expression for the speed of the wave. **(3 marks)** *April 2022*

5 What do you mean by a monochromatic plane wave ? Give its general form. **(3 marks)** *April 2022*

***Poynting's theorem and the Poynting vector***

6 Define Poynting vector and give an expression for the same. **(2 marks)** *November 2021*

7 What is Poynting vector ? Give an expression for the same. **(3 marks)** *April 2022*

***Energy and radiation pressure of electromagnetic waves***

8 The average power per unit area transported by an electromagnetic wave is called its _____. **(1 mark)** *November 2019*

9 The average power per unit area transported by an electromagnetic wave is called its _____. **(1 mark)** *November 2021*

***Reflection and transmission at a boundary***

10 The velocity of electromagnetic waves in a medium with permittivity $\varepsilon$ and permeability $\mu$ is given by _____. **(1 mark)** *November 2019*

## Section B: 4 to 5 marks

***Electromagnetic waves in vacuum***

11 Write down the real electric and magnetic field for a monochromatic plane wave of amplitude $E_0$, frequency $\omega$ and phase angle zero that is travelling in the positive $x$ direction and polarized in the $z$ direction. **(4 marks)** *November 2019*

12 From Maxwell's equations find the wave equation for the electric field in vacuum. **(4 marks)** *November 2019*

***Poynting's theorem and the Poynting vector***

13 Prove that for a plane monochromatic wave, the Poynting vector is the energy density times the velocity of the wave. **(5 marks)** *April 2022*

***Energy and radiation pressure of electromagnetic waves***

14 What is radiation pressure due to electromagnetic field? Find how it is related to intensity of the wave. **(4 marks)** *November 2019*

***Reflection and transmission at a boundary***

15 Give the fundamental laws of geometrical optics considering the reflection and transmission of electromagnetic waves at a boundary separating two linear media. **(5 marks)** *April 2022*

## Section C: 10 and 11 marks

***Waves in one dimension and the wave equation***

16 Find the boundary conditions for reflection and transmission of electromagnetic wave, when propagated through a string which is tied onto a second string. **(10 marks)** *November 2019*

***Electromagnetic waves in vacuum***

17 Obtain the wave equation for the $\vec{E}$ and $\vec{B}$ vectors in free space. Using a plane wave solution show that the electromagnetic waves are transverse in nature and the $\vec{E}$ and $\vec{B}$ vectors are in phase and mutually perpendicular. **(11 marks)** *April 2022*

***Reflection and transmission at a boundary***

18 Find expressions for the transmission and reflection coefficients when a plane polarized monochromatic wave of angular frequency $\omega$ passes normally through the boundary between two linear dielectrics. **(10 marks)** *November 2021*